Names: ____________________ ____________________ Table: _____

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Discussion Assignment 2

Getting Ready for the Winter Olympics MATH 2210, Spring 2018

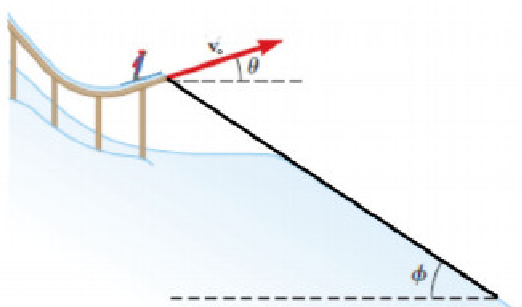
Background

Any time that an object is launched into the air with a given velocity and launch angle, the path the object travels is determined almost exclusively by the force of gravity. Whether in sports such as archery or ski jumping, in military applications with artillery, or in fields like firefighting, it is important to be able to know when and where a launched projectile will land. We can use our knowledge of vector-valued functions in order to completely determine the path traveled by an object that is launched from a given position at a given angle from the horizontal with a given initial velocity.

Assume we fire a projectile from a launcher and the only force acting on the fired object is the force of gravity pulling down on the object. That is, we assume no effect due to spin, wind, or air resistance. With these assumptions, the motion of the object will be planar, so we can also assume that the motion occurs in two-dimensional space. Suppose we launch the object from an initial position (x_o, y_o) at an angle θ with the positive x-axis, and that we fire the object with an initial speed of $v_0 = |\boldsymbol{v}(0)|$, where $\boldsymbol{v}(t)$ is the velocity vector of the object at time t. Assume g is the positive constant acceleration force due to gravity, which acts to pull the fired object toward the ground (in the negative y-direction). Note particularly that there is no external force acting on the object to move it in the x direction.

Question

Suppose a skier with mass 70 kg leaves a ski jump 3 m above the snow line below with an initial velocity of $|\boldsymbol{v}_o| = 30\,\mathrm{m/s}$, $\theta = 12^\circ$ above the horizontal, and that the slope is inclined at $\phi = 40^\circ$.



What is the distance from the edge of the ramp to where the jumper lands? Choose the best answer form the selection below.

(a) 74 m

(b) 185 m

(c) 241 m

(d) 384 m

Report

You are to write a report, two pages maximum, that convinces an audience of your peers that the skier lands at the distance from the edge that you have claimed. This report must be typed, saved as a pdf and submitted to your discussion section's WyoCourse by the start of next weeks discussion period. You may use any text editor you prefer, but appropriate typesetting for all equations and notations must be used. A scoring rubric, guideline on writing good mathematics, and example reports are available on WyoCourses.

As part of your report you must derive the position vector function, $\boldsymbol{r}(t)$, for the given projectile:

- First observe that since gravity only acts in the downward direction and that the acceleration due to gravity is constant, the acceleration vector is $\boldsymbol{a}(t) = \langle 0, -g \rangle$.
- Find all velocity vectors for the given acceleration vector $\boldsymbol{a}(t)$. When you anti-differentiate, remember that there is an arbitrary constant that arises in each component.
- Use the given information about initial speed and launch angle to find $\boldsymbol{v}_o$, the initial velocity of the projectile. You will want to write the velocity vector in terms of its components, which will involve $\sin(\theta)$ and $\cos(\theta)$.
- Next, find the specific velocity function $\boldsymbol{v}(t)$ for the projectile. That is, combine your work in the first and second parts to determine expressions in terms of v_o and θ for the constants that arose when integrating.
- Find all possible position vectors for the velocity vector $\boldsymbol{v}(t)$ by anti-differentiating.
- Use the fact that the object is fired from the position (x_o, y_o) to show that

$$\boldsymbol{r}(t) = \left\langle x_o + v_o \cos(\theta)t,\ y_o + v_o \sin(\theta)t - \frac{1}{2}gt^2 \right\rangle.$$